

BURSTY BULK FLOW IONOSPHERIC IMPACTS FROM TRIPLE-CONJUNCTIVE OBSERVATIONS

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George (2026); Annales Geophysicae, under review

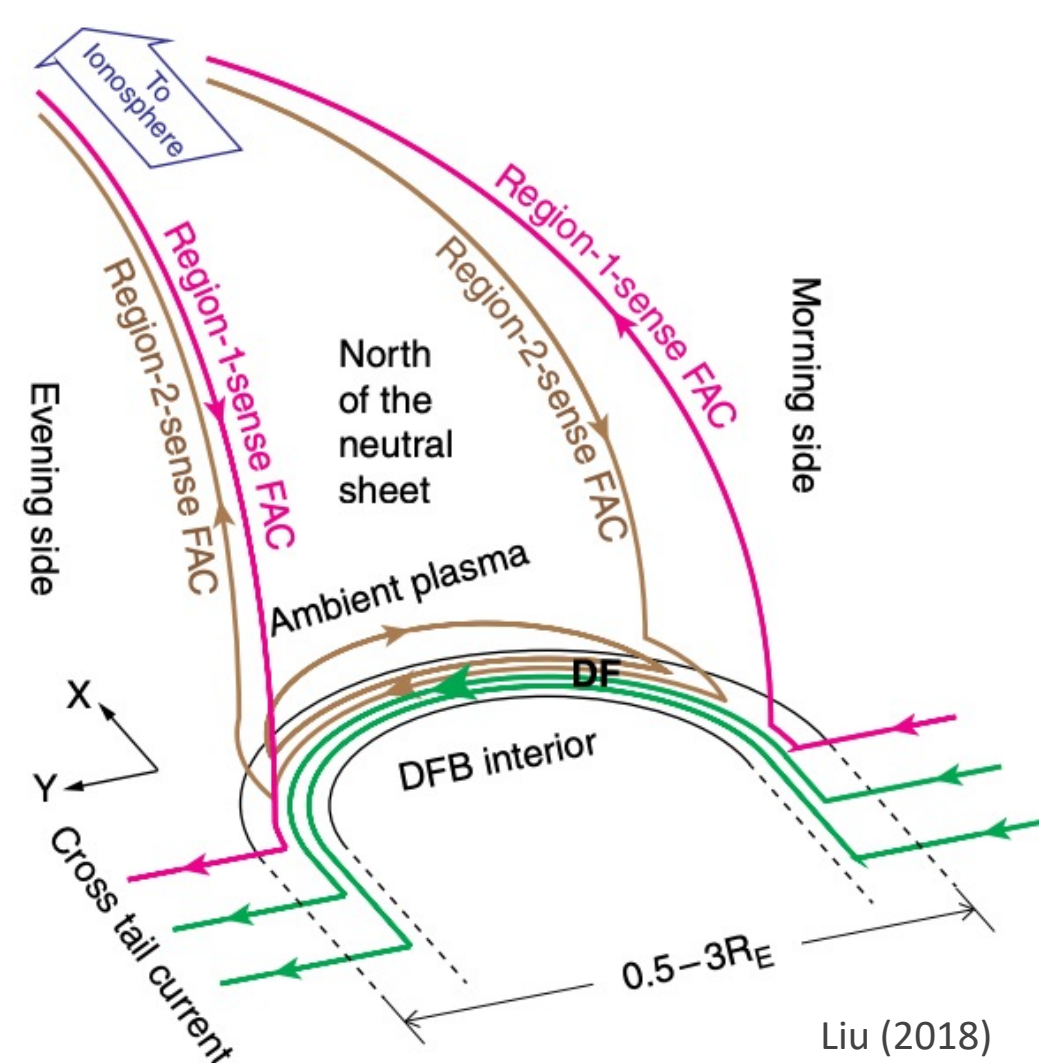


Identified magnetic conjunctions between MMS and THEMIS **bursty bulk flow** detections, **Swarm** satellites and ground-based **radar** → 11 unique triple conjunctions from ~ 11,000 BBFs → detailed analysis of **case study** to evaluate ionospheric impacts of BBF

BACKGROUND

- BBFs are fast, mesoscale ion flows in magnetotail
- Couple to ionosphere via field aligned currents (FAC)
- Localised ionospheric disturbances at BBF magnetic footprint

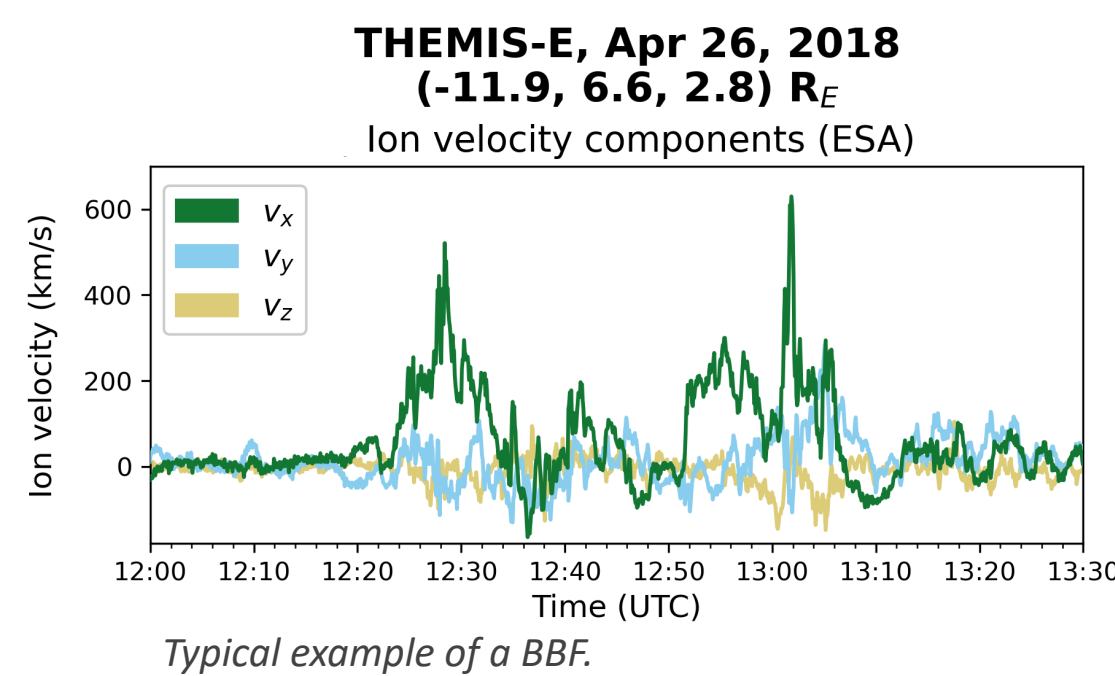
Role of mesoscale phenomena (including BBFs) in cross-region coupling is an open question.



TRIPLE CONJUNCTION IDENTIFICATION

Bursty bulk flow: THEMIS and MMS databases

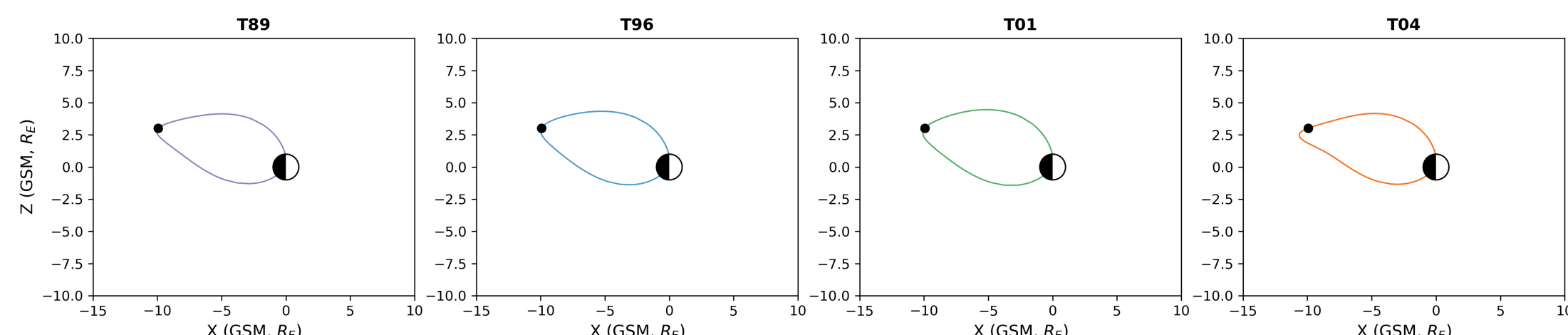
- THEMIS: 8697 BBFs; 2017 – 2024; George (2026); submitted to JGR)
- MMS: 2394 BBFs; 2017 – 2021; (Richard, 2022)



Field aligned currents: Swarm constellation in low Earth orbit.

Ionospheric response: Radar stations - Poker Flat (PKR), Tromso (TRO), Svalbard (ESR)

BBF magnetic footprint: magnetic field lined traced from observed location in magnetotail to 110 km altitude with T89, T96, T01 and T04 models then averaged across magnetic field models.



BBF / radar conjunction

Swarm / radar conjunction

BBF / Swarm / radar conjunction

- BBF footprint and Swarm conjugate with same radar station
- Swarm conjunction occurs within 10 minutes of BBF detection

13 triple conjunctions identified (11 unique)

21/6/17; MMS, Swarm-B, TRO
31/1/18; TH-A, Swarm-B, PKR
10/8/18; MMS, Swarm-B, TRO
18/7/21; TH-A, Swarm-B, PKR
1/9/21; MMS, Swarm-B, PKR
27/9/22; TH-D, Swarm-A, TRO
3/11/23; TH-D, Swarm-A, PKR

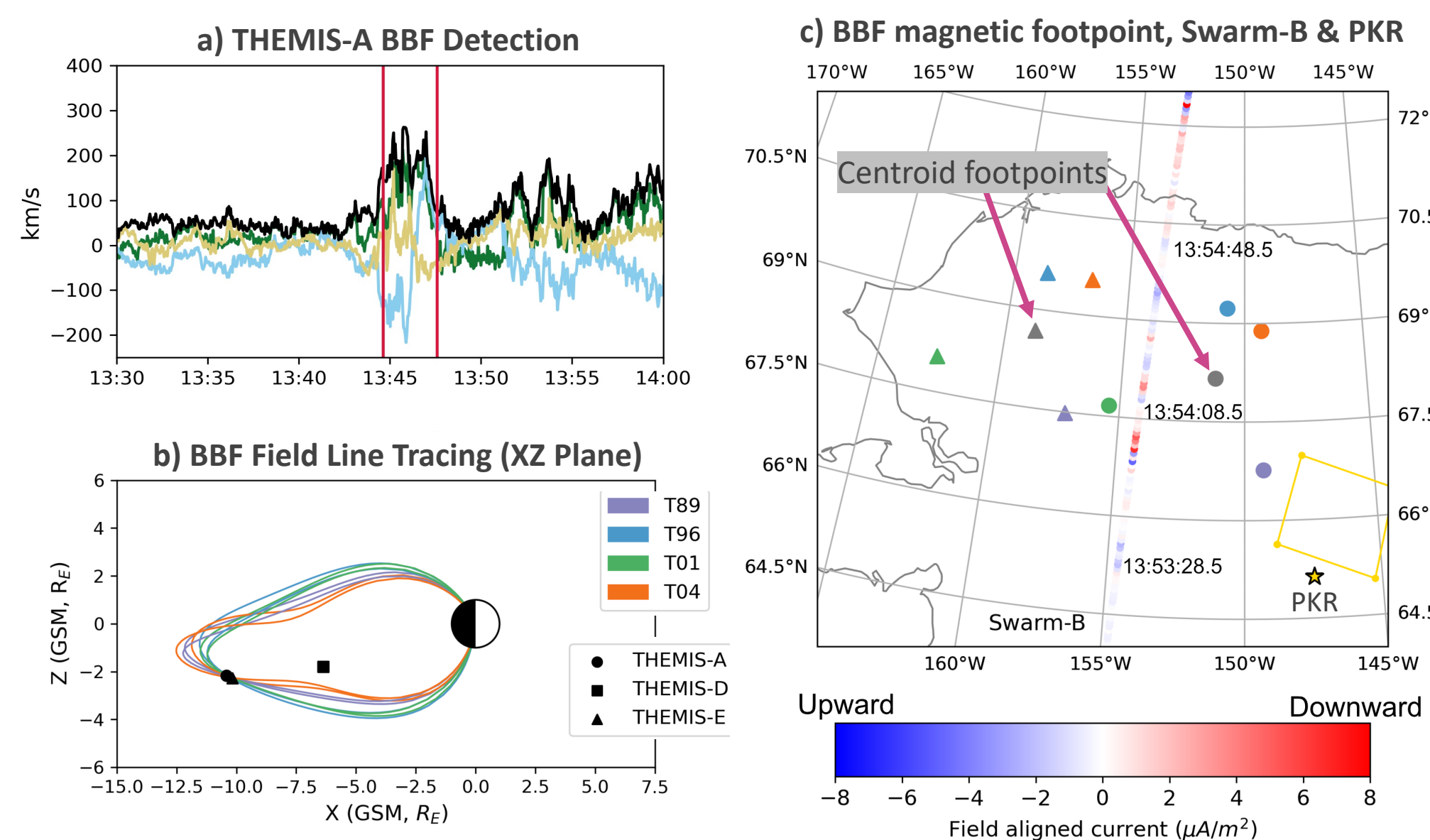
3/11/23; TH-E, Swarm-A, PKR
11/11/23; TH-D, Swarm-A, PKR
11/11/23; TH-E, Swarm-A, PKR
29/11/23; TH-E, Swarm-A, PKR
7/12/23; TH-D, Swarm-A, TRO
15/12/23; TH-D, Swarm-A, TRO

CASE STUDY: JAN 31, 2018

BBF DETECTION & MAGNETIC FOOTPOINT

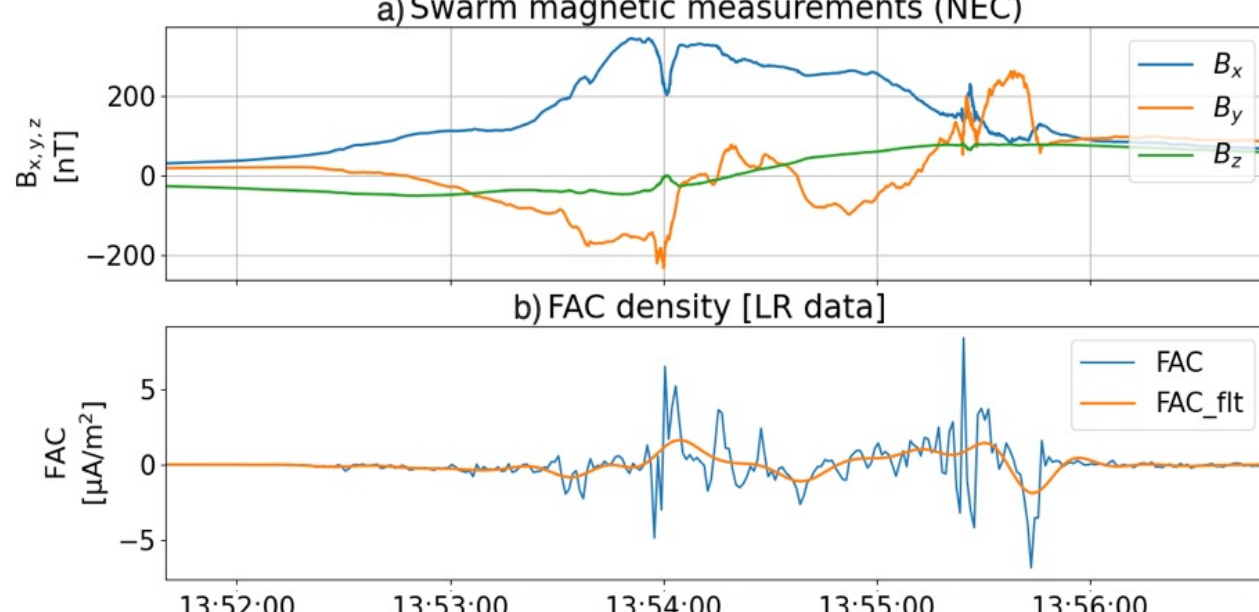
Triple conjunction between magnetic footprint of BBF detected by **THEMIS-A**, **Swarm-B** and **Poker Flats** radar station.

BBF also detected by THEMIS-E (TH-D in lobe). Footpoint in Alaska. Occurred in substorm recovery phase during quiet driving conditions.



FIELD ALIGNED CURRENT SIGNATURE

Swarm-B field aligned current observations



Swarm-B detected FAC associated with BBF at 13:54 when overpassing magnetic footprint.

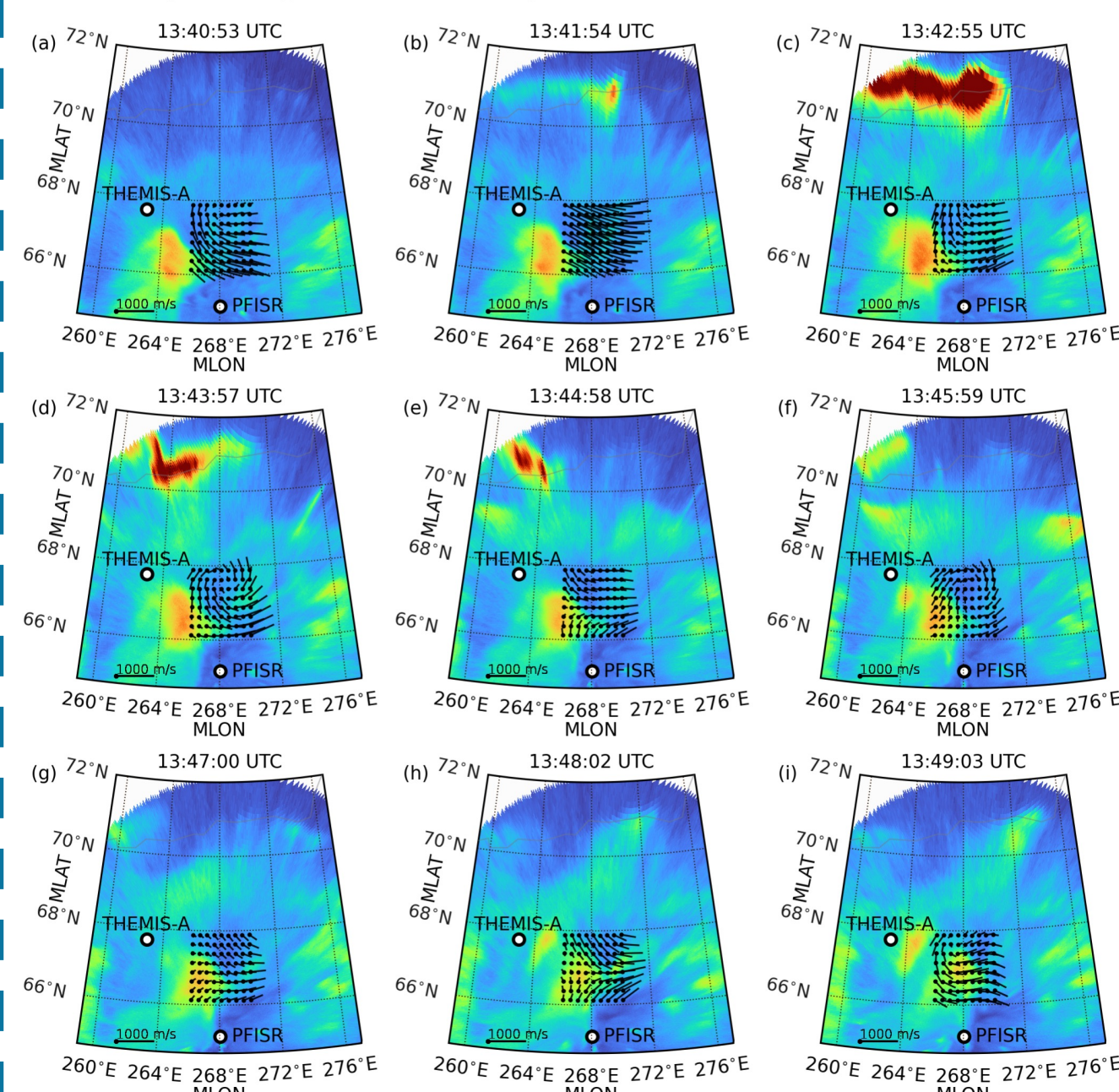
Multiscale multivariate analysis of FAC showed low planarity, high inclination and ~ 100 km scale size.

IONOSPHERE RESPONSIC

Auroral streamer + FAC signatures in ionospheric plasma vortices.

Ionospheric plasma variations throughout substorm. Further research needed to isolate BBF effects from substorm dynamics.

Ionospheric plasma velocity vectors and auroral emissions



Poker Flat Incoherent Scatter Radar observations: Beam 2

